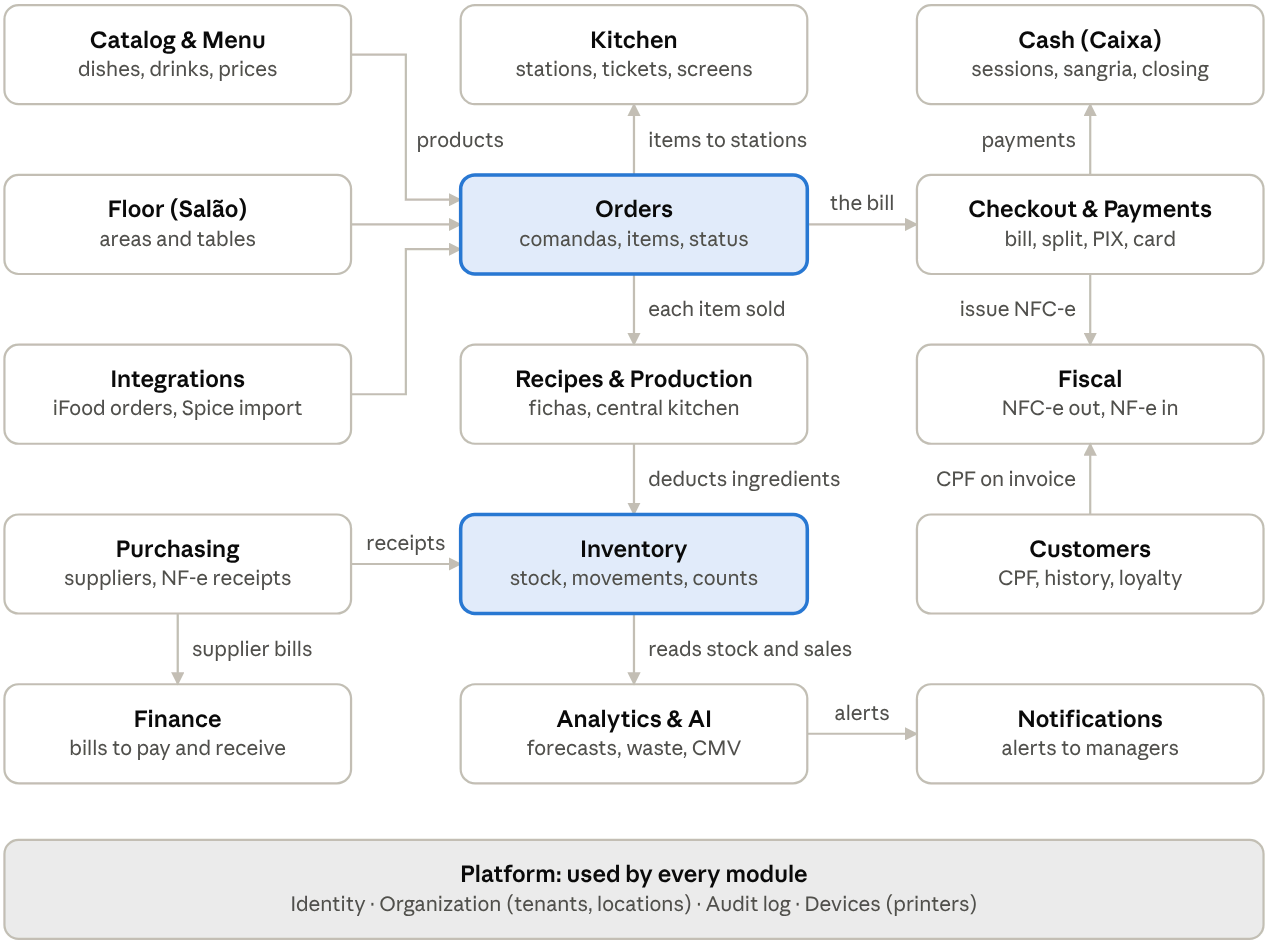
Restaurant System — Module Map

Oct 14, 2026 · @Igor Mori

Overview

The system has 19 modules in 5 domains. Orders and Inventory are the hubs: every sale passes through Orders, and every stock change through Inventory.

Orders and Inventory are the two hubs the other modules connect to



module map · 19 modules, 2 hubs

Not drawn, to keep it readable: Fiscal also brings supplier invoices into Purchasing, Finance also receives card payments and cash closings, and Customers attach to Orders.

What was missing

Your list covers the salão well. The biggest gaps are recipes, stock movements and the central kitchen, which are exactly what the AI features depend on.

Missing piece	Why it matters	Goes in
Recipes (fichas técnicas)	Link each dish to its ingredients in grams; without them there is no CMV (cost of goods sold) or waste detection	Recipes & Production
Stock movements	Every stock change is a record: purchase, sale, waste, transfer, count adjustment	Inventory
Units and conversions	Buy a box of 12, store single units, use grams in recipes	Inventory
Inventory counts (contagem)	Physical count vs system; the gap is unexplained loss	Inventory
Waste log (registro de perdas)	Expired, dropped or burned items, so known loss is separated from unknown loss	Inventory
Lots and expiry dates (validade)	Near-expiry alerts and promotion suggestions	Inventory
Packaging (embalagens)	Delivery boxes and cups are a real cost; treat them as stock items	Inventory
Central kitchen production	Turn raw ingredients into sauces and portions, then transfer them to each store	Recipes & Production
Modifiers (adicionais)	"No onion", stuffed crust, doneness: they change the price, the stock used and the kitchen ticket	Catalog & Menu
Kitchen stations (praças)	Send each item to the right printer or screen: kitchen, bar, pizza oven	Kitchen
Payment details	PIX, card, cash, meal vouchers (vale-refeição), split bill, 10% service charge, discounts	Checkout & Payments

Missing piece	Why it matters	Goes in
Cancellations with approval	A reason and manager approval on every cancelled item; the main anti-fraud control	Orders + Audit
Fiscal setup	Tax codes per product (NCM, CFOP, CST), tax regime, digital certificate, NFC-e series per CNPJ	Fiscal
Bills to pay and receive	Supplier bills with due dates; card payments that arrive later, minus fees	Finance
Audit log	Who cancelled, discounted or opened the cash drawer, and when	Platform
Printers and devices	Which printer serves which station and which cash register	Platform

NCM, CFOP and CST are the tax classification codes every product needs on an invoice.

Modules by domain

19 modules in 5 domains. Each can become its own `RestaurantManagement.Modules.<Module>` project; small ones can share a project at first.

Platform

Module	Status	Main tables	Connects to
Identity	Exists	Users, Roles, Permissions, RefreshTokens	Every module (who did what)
Organization	Exists	Organizations (tenants), Locations (store or central kitchen), Settings, FiscalProfiles	Every module: rows carry OrganizationId, most also LocationId
Audit	New	AuditLog (who, what, when, before and after)	Orders, Checkout, Cash and Inventory write to it
Devices	New	Printers, Terminals, StationPrinters	Kitchen, Checkout, Cash

Front of house (salão)

Module	Status	Main tables	Connects to
Catalog & Menu	Partly exists (Menu, MenuCategories)	Products, ModifierGroups, Modifiers, Menus, MenuCategories, Prices (per location and channel)	Orders, Recipes, Inventory (resale items), Fiscal (tax codes)
Floor (Salão)	New	Areas (Floor 1, Inside, Outside), Tables	Orders
Orders	New	Orders (table, comanda, takeout, delivery), OrderItems, ItemModifiers, Cancellations, TableTransfers	Floor, Catalog, Kitchen, Checkout, Recipes, Customers
Kitchen	New	ProductionStations (kitchen, bar, oven), StationRouting, KitchenTickets	Orders, Devices

Module	Status	Main tables	Connects to
Customers	New	Customers (name, phone, CPF), Consents (LGPD), loyalty later	Orders, Fiscal (CPF on the invoice)

Money

Module	Status	Main tables	Connects to
Checkout & Payments	New	Bills, BillSplits, Discounts, ServiceCharges, Payments (PIX, credit, debit, cash, meal voucher), Tips	Orders, Cash, Fiscal, Finance
Cash (Caixa)	New	CashRegisters, CashSessions (opening balance, expected vs counted), CashMovements (sales, sangria, suprimento)	Checkout, Finance
Fiscal	New	NfceDocuments, FiscalSeries, ProductTaxData, InboundNfe (supplier XMLs)	Checkout, Purchasing, Organization (CNPJ, certificate)
Finance	New	AccountsPayable, AccountsReceivable, Expenses, CardFees, IncomeStatement (DRE)	Purchasing, Checkout, Cash

Sangria is cash taken out of the drawer, suprimento is cash added, and DRE is the income statement.

Back of house

Module	Status	Main tables	Connects to
Inventory	New	StockItems (ingredients, resale items, packaging), Units, UnitConversions, StockLevels, StockMovements, InventoryCounts, WasteLog, Lots	Purchasing, Recipes, Orders, Analytics
Recipes & Production	New	Recipes, RecipeItems, ProductionOrders, Transfers (central kitchen to stores)	Catalog, Inventory, Organization (the central kitchen is a Location)
Purchasing	New	Suppliers, SupplierItems (their code to your stock item), PurchaseOrders, PurchaseReceipts, PriceHistory	Inventory, Finance, Fiscal

Intelligence and integrations

Module	Status	Main tables	Connects to
Analytics & AI	New	Forecasts, Alerts, Recommendations, WeeklyReports	Reads Orders, Inventory, Purchasing, Recipes
Notifications	New	NotificationRules, Notifications	Analytics, Inventory, Cash
Integrations	New	IntegrationAccounts, ExternalOrders (iFood), ImportJobs (Spice)	Orders, Catalog

Later, not in the first build: Staff and shifts, Reservations, and Subscription billing for when you sell to other restaurants.

Key business flows

Four flows cross most modules; if these work end to end, the system works.

1. Order to cash (salão)

1. The waiter opens an order on a table or comanda (Floor, Orders).
2. Each item goes to its station: kitchen, bar or oven (Kitchen).
3. The customer asks for the bill; split, service charge and discounts are applied (Checkout & Payments).
4. Payments are recorded by method in the open cash session (Cash).
5. The cashier issues the NFC-e when needed (Fiscal).
6. Stock is deducted using each dish's recipe (Recipes & Production, Inventory).
7. The sale feeds reports and forecasts (Analytics & AI).

2. Purchase to stock

1. A supplier invoice arrives: the NF-e XML is fetched automatically, or Claude reads a photo of a paper note (Fiscal, Purchasing).
2. Supplier item codes are matched to your stock items, and prices are updated (Purchasing).
3. Stock goes up at the receiving location (Inventory).
4. A bill to pay is created with its due date (Finance).
5. Price increases are flagged (Analytics & AI).

3. Central kitchen to stores

1. A store falls below its target stock, or the forecast predicts higher demand (Inventory, Analytics & AI).
2. A replenishment request goes to the central kitchen (Recipes & Production).
3. The central kitchen produces: ingredients out, prepared items in, each batch with a lot and expiry date (Recipes & Production, Inventory).
4. The transfer is sent and received, and stock moves between locations (Inventory).
5. Items close to expiry are flagged for a promotion (Analytics & AI).

4. Cash close (fechamento)

1. The cashier opens a session with the opening balance (Cash).
2. Sales, withdrawals (sangria) and additions (suprimento) are recorded during the shift.
3. At close, the cashier counts each payment method; the system compares expected vs counted and records the difference.
4. Several sessions per day are allowed, one per shift or cashier; the day's totals feed Finance.

Key design decisions

Decided and locked; changing them after real data exists is expensive.

Dishes and beverages are both Products. A Product is what you sell; a Stock item is what you buy and store. They connect through a recipe or a one-to-one link:

Example	Product type	How it uses stock
Margherita pizza	Prepared	Recipe: flour, mozzarella, tomato sauce
Caipirinha	Prepared (bar)	Recipe: cachaça, lime, sugar
Coca-Cola can 350 ml	Resale	1 can of the matching stock item
Lunch combo	Combo	The products inside it

Catalog and Menu are separate. The catalog defines each product once; menus decide where it appears (salão, iFood, lunch), and prices can differ per location and channel. Your current MenuCategories belong on the menu side.

Every table is tenant-scoped. Rows carry OrganizationId, and most carry LocationId; EF Core Global Query Filters apply the filter to every query.

The central kitchen is a Location. Its type is "production", so stock, transfers and costs work exactly as they do in the stores.

Fiscal settings live on the Location. Mercado Municipal and Batel each have their own CNPJ, so each needs its own digital certificate and NFC-e series.

Stock is a ledger of movements. Never edit a stock level directly: every change is a movement, and the level is their sum. That is what makes waste detection and audits possible.

Sensitive actions leave a trail. Money uses `decimal`; cancellations and discounts need a reason and a permission, and all of them go to the Audit log.

Build phases

Spice stays live until phases 1 and 2 are both done; the AI in phase 3 then builds on the stock data from phase 2.

Phase	Goal	Modules
0. Foundations (now)	A secure base every feature inherits	Identity, Organization, Audit, tenant filters
1. Run the salão	Build the salão operation, with Spice still live	Catalog & Menu, Floor, Orders, Kitchen, Checkout & Payments, Cash, Fiscal, Devices
2. Control stock	Know your stock and dish costs, then switch off Spice	Inventory, Recipes & Production, Purchasing
3. Intelligence	Forecasts, waste alerts and central kitchen replenishment	Analytics & AI, Notifications, Integrations (Spice history import)
4. Grow	Ready to sell to other restaurants	Finance, Customers and loyalty, iFood, Staff, Reservations, Subscription billing

We keep Spice live until phase 2 is complete, then switch over one unit at a time.